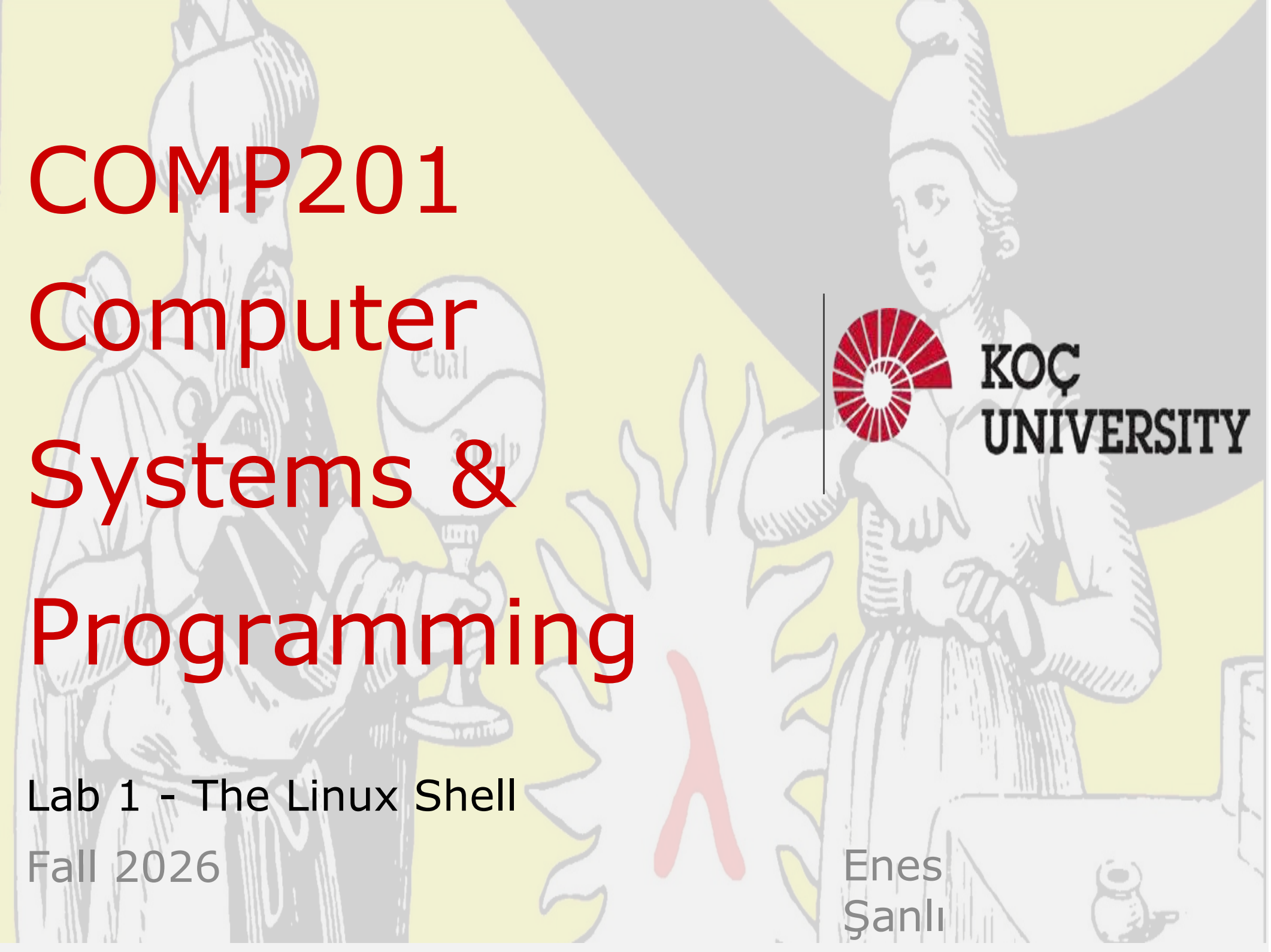


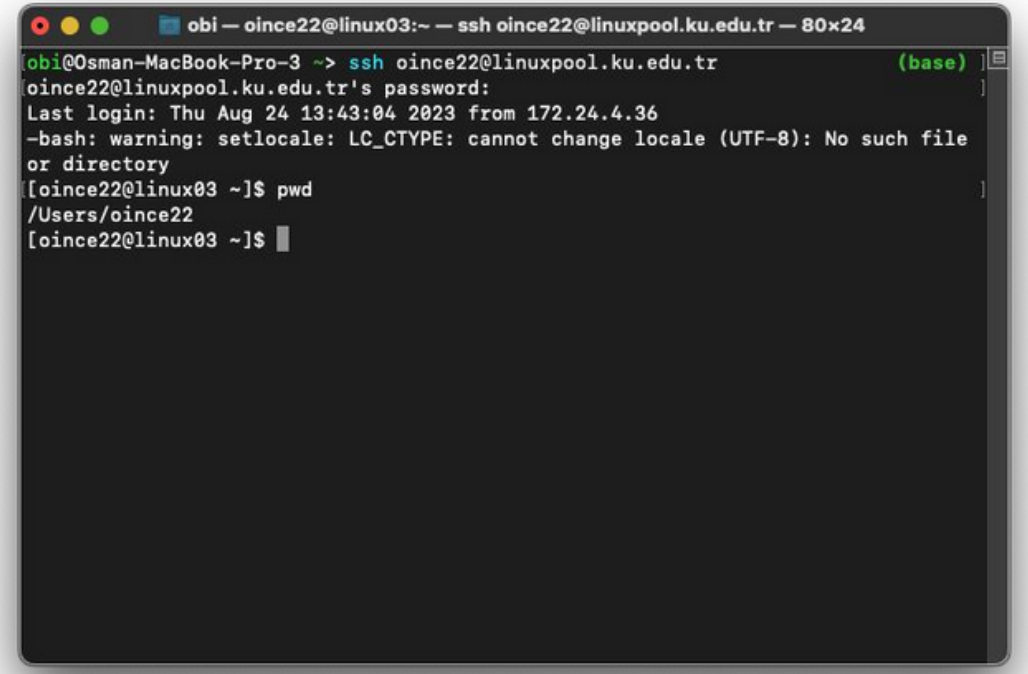
COMP201 Computer Systems & Programming

Lab 1 - The Linux Shell

Fall 2026

Enes Şanlı

What is shell?



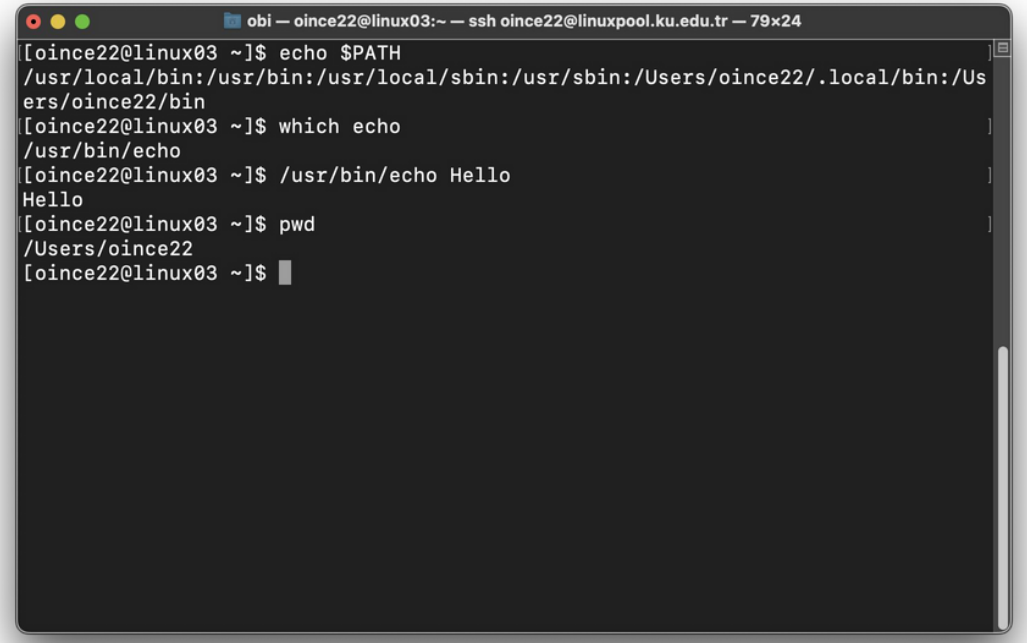
- Linux shell is the interface between you and OS that controls hardware.
- The most commonly used shell is called BASH – Bourne Again Shell
  - The default shell in Linuxpool
- username@hostname:curr_dir$
  - username: oince22
  - hostname: linux03
  - curr_dir: /Users/oince22

How to connect?

ssh USERNAME@linuxpool.ku.edu.tr

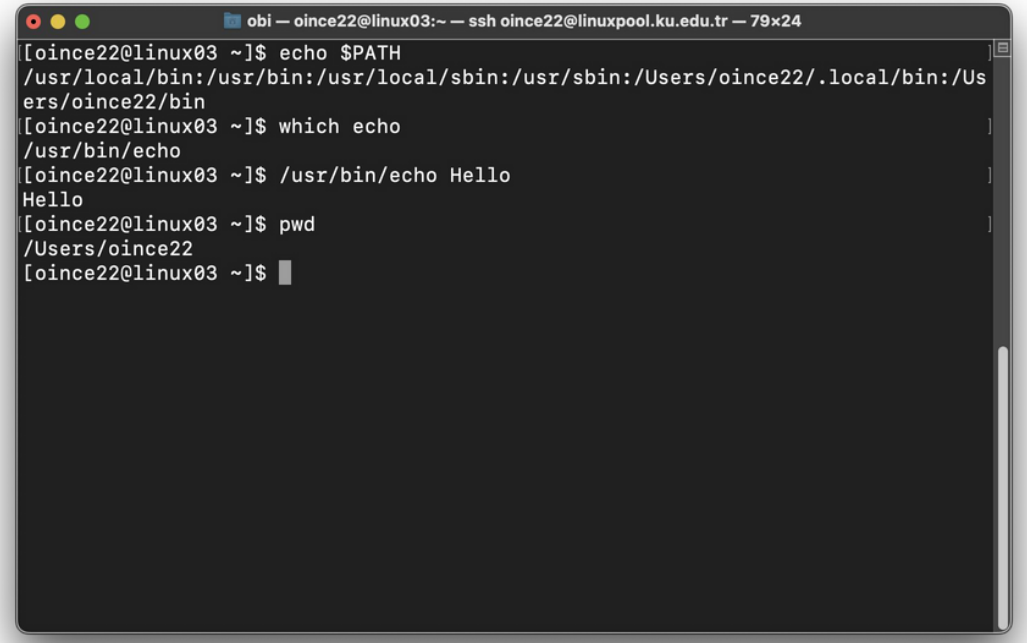
1. Type your password when prompted.
2. If you see a warning about SSH host keys, click or enter “yes.”

Executing system programs



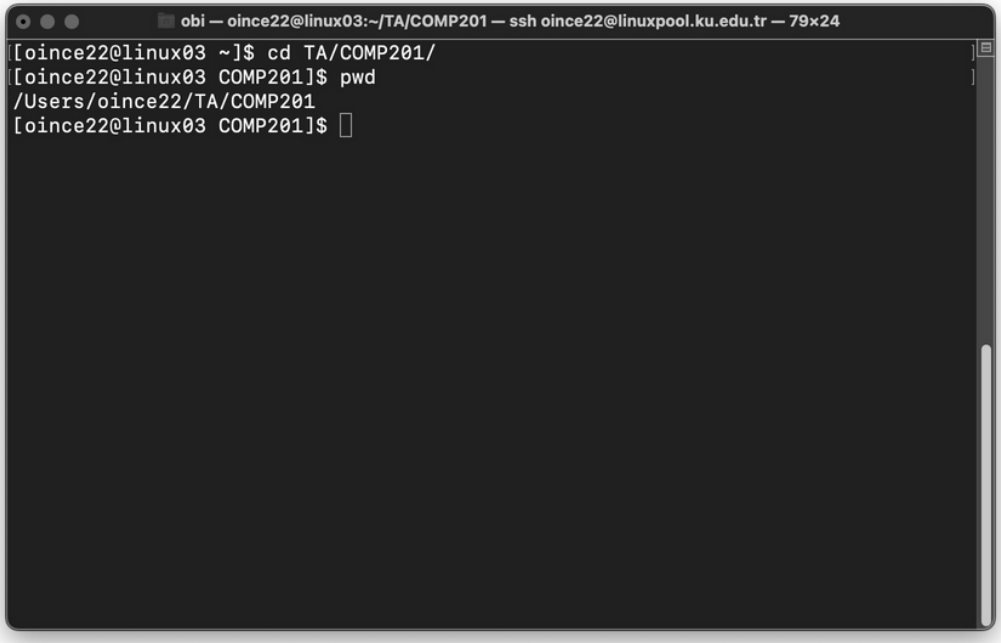
- Execute programs
- date
  - This program prints current date and time
- echo
  - This program prints the input argument

Path and $PATH



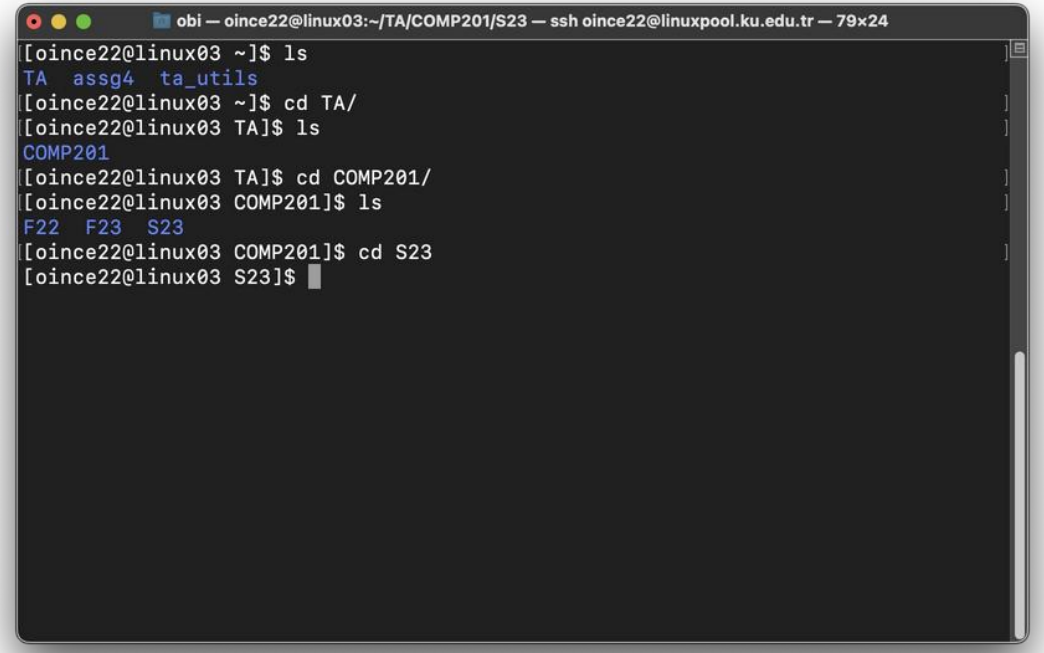
- $PATH
  - A variable that contains addresses where system look for programs to execute
- which
  - Prints which file is being executed given an input program name
- pwd
  - This program prints current working directory
  - Stands for “print working directory”

Path



- cd
  - Changes the working directory
  - .. is the parent directory
  - . is the current directory
  - Tilda (~) is the /Users/<username> directory
    - This is true in Linuxpool
    - May be different in another machine
- Absolute vs relative path
  - Relative: TA/COMP201from ~ (home)
  - Absolute: /Users/oince22/TA/COMP201

Listing files and directories



- ls
  - Prints files and directories under current working directory

Flags with Commands in Linux

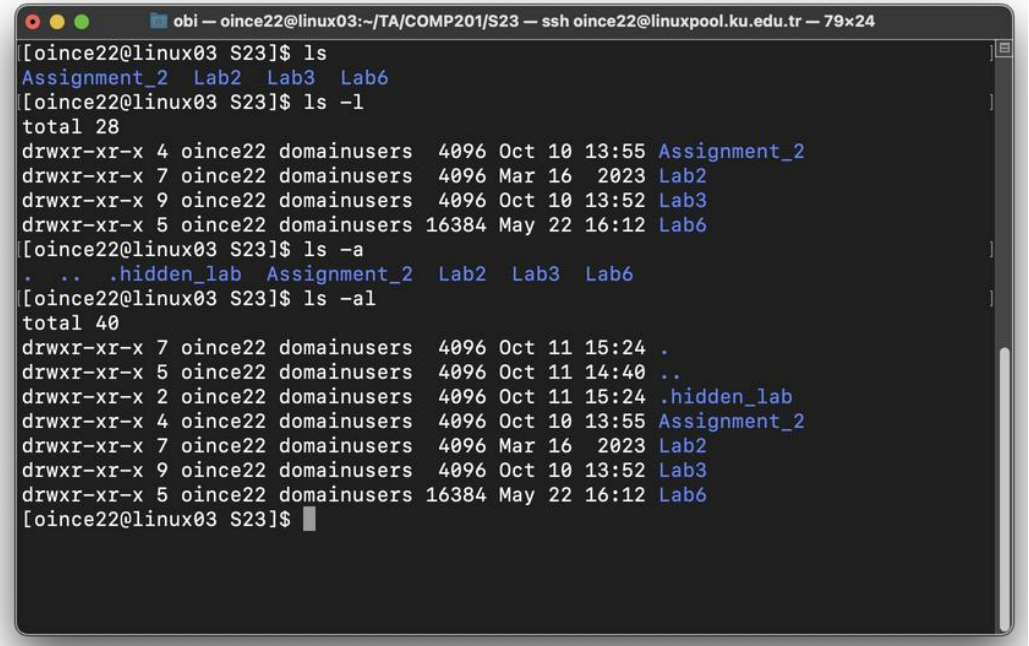
- Many Linux commands have flags that can be used to modify their behavior.
- Flags are usually preceded by one or two dashes, followed by a letter or a word.

  Flags can be used to:

  - Control the output of a command
  - Specify a file or directory to work with
  - Modify the command's behavior in other ways

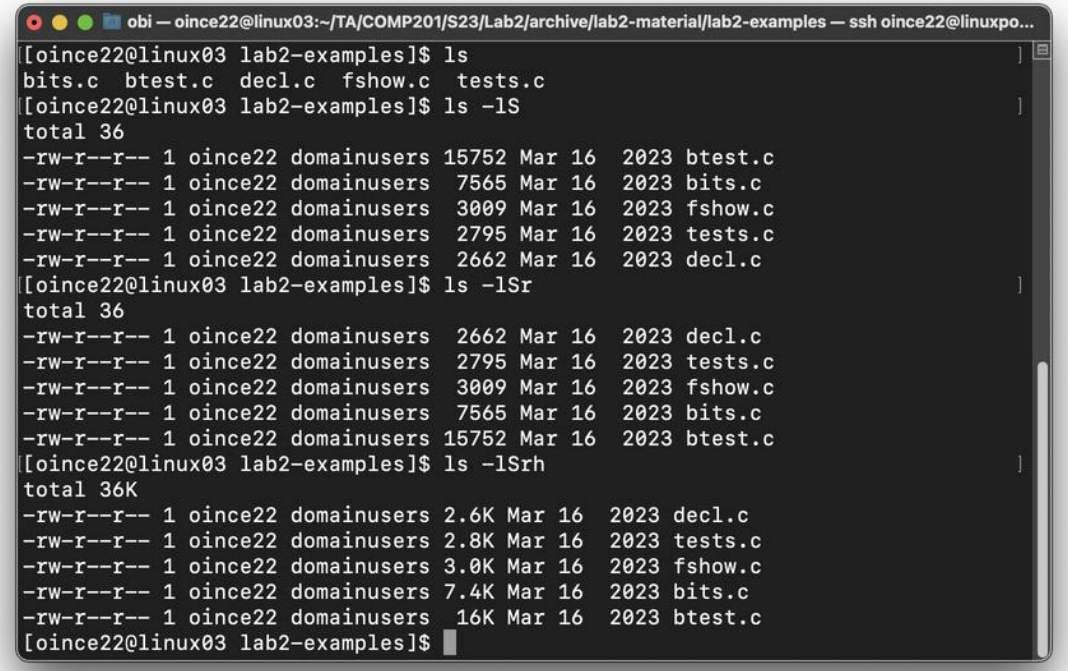
Flags with Commands in Linux

- Let's look at an example: ls command. By
- default, it lists contents of the current folder.
- But we can use flags to modify its behavior.
- For example,
  - -l flag to list the contents of the directory line-by-line, long-format including additional info about file permissions, owner, and size.
  - -a flag to display all files, including hidden files (usually not displayed by default).
- To use both flags together, type ls -la
  - Combine as many as you want!



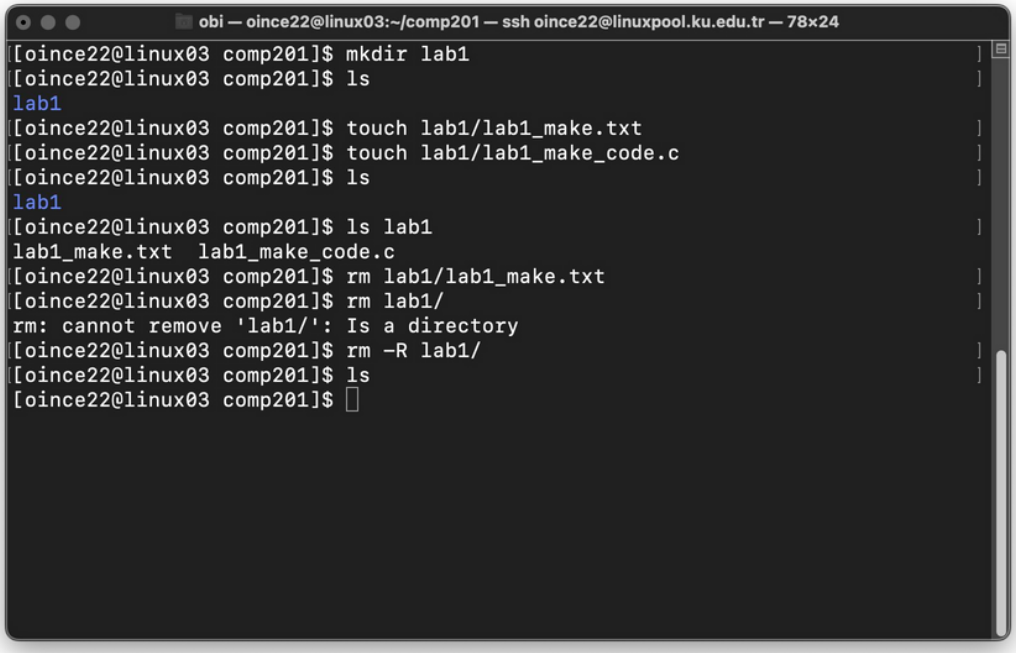
To learn more about the flags available for a command, type man command
To learn details about the ls command and its flags → man ls

Listing files and directories



- You can use -S flag to display files sorted by their sizes, and -r option for reverse sorting. You can use -h flag to display file sizes in a human-readable format.

Making/Removing folders and files



- mkdir <folder_name>
  - Makes a new directory in the given working directory with the given "folder_name".
- touch
  - Creates a file with desired extension and name
- rm
  - Removes a file or folder.
  - For removing folders you need to use -R option

Chmod

- Chmod (short for "change mode") is a command in Linux that allows users to change the read, write, and execute permissions of files and directories.
- The syntax for chmod is as follows:
  - chmod [options] MODE FILENAME
- The mode is a combination of the letters "r" (read), "w" (write), and "x" (execute).
- Permissions can be granted to three different user groups:
  - The fileowner
  - The group owner
  - All users

File Permission in Linux

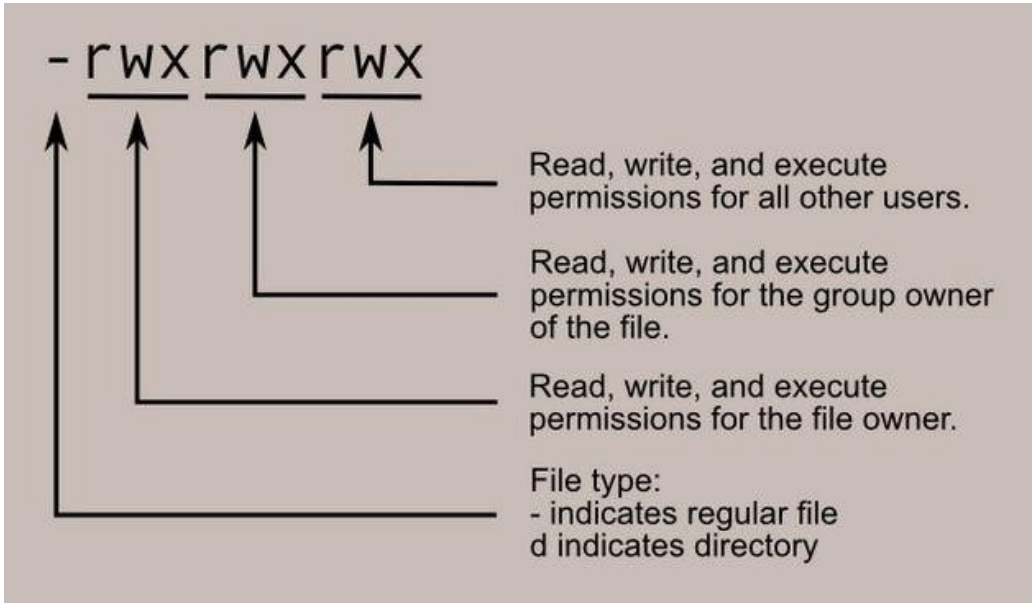


Image source: http://linuxcommand.org/lc3_lts0090.php

File Permission in Linux

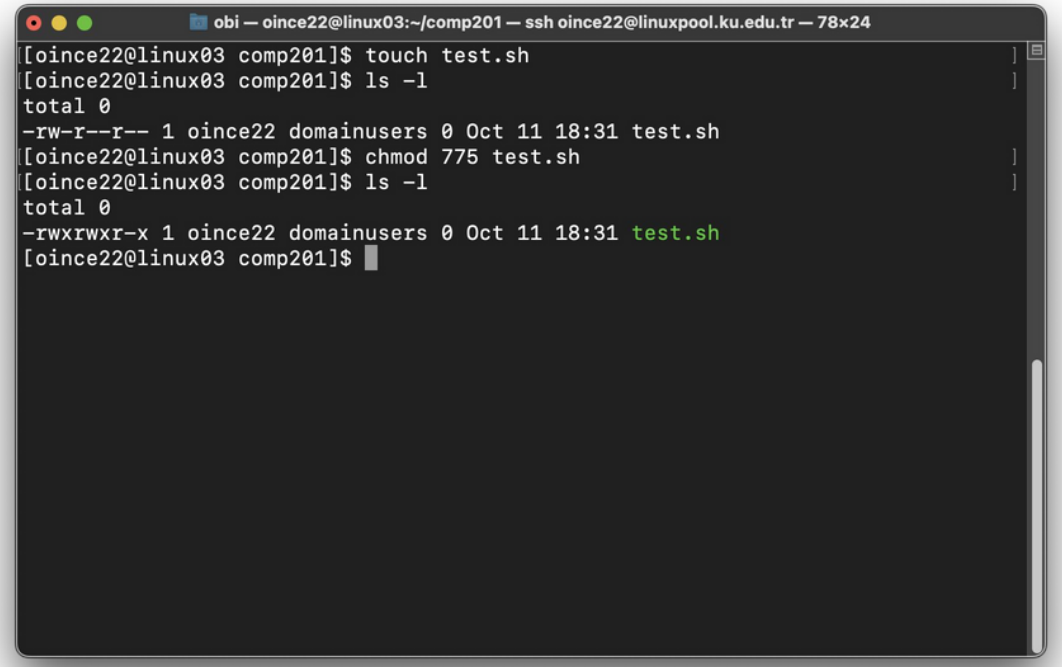
```
rwx rwx rwx = 111 111 111
rw- rw- rw- = 110 110 110
rwx --- --- = 111 000 000

and so on...

rwx = 111 in binary = 7
rw- = 110 in binary = 6
r-x = 101 in binary = 5
r-- = 100 in binary = 4
```

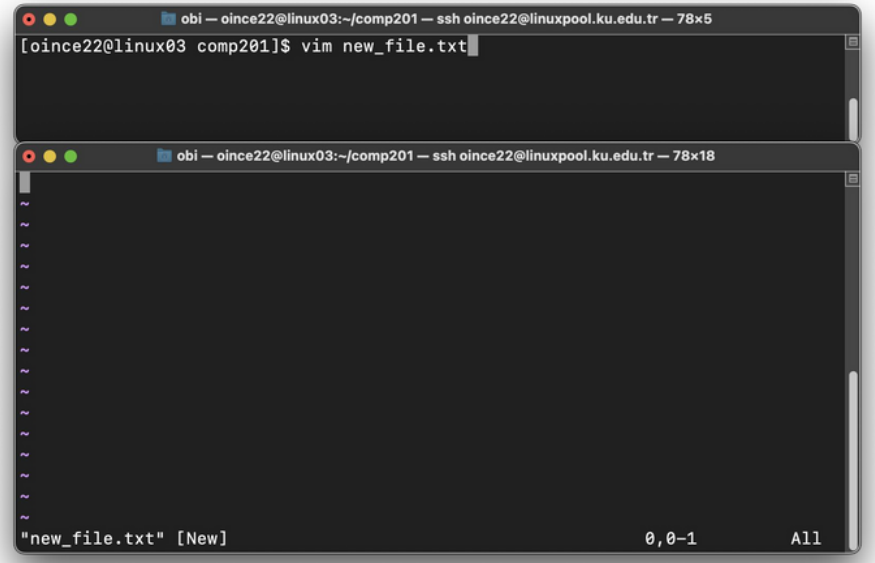
Image source: http://linuxcommand.org/lc3_lts0090.php

File Permission in Linux



Initially, test.sh cannot be executed, to grant -rwx rwx r-x permission to test.sh file execute chmod 775 test.sh command.

What is Vim?



- Vim is the default text editor in the UNIX operating system.
- Using vim, we can create a new file, read, and edit an existing file.
- To open vim, type vim or vim FNAME. If the file FNAME doesn't exist, it will be created when you save it.

Operation Modes in Vim





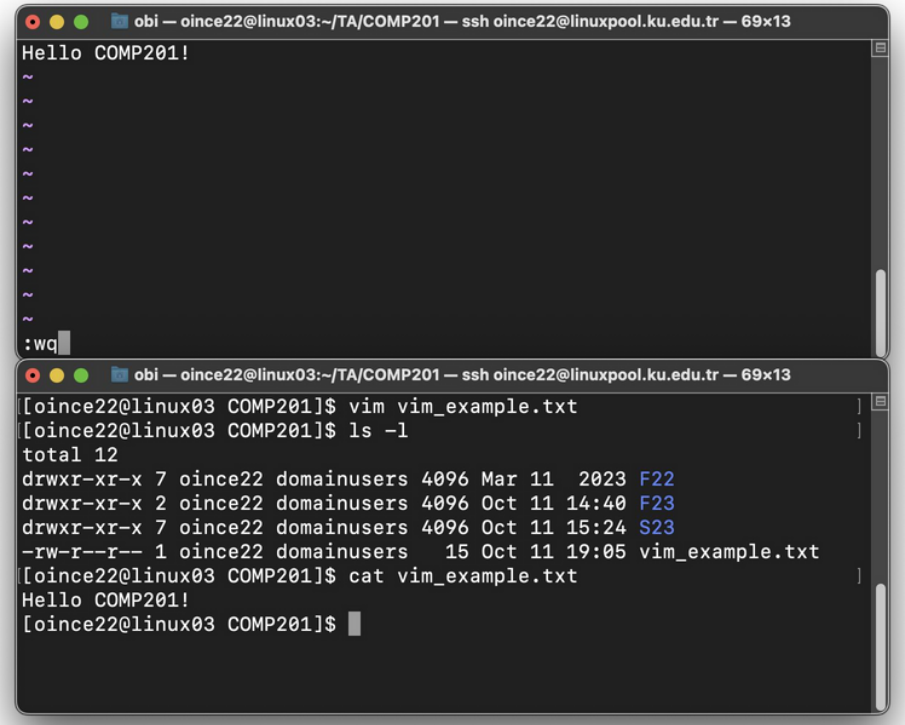
Normal mode

- Thedefaultmode in vim.
- Every character you type is interpreted as a command.

Insert mode

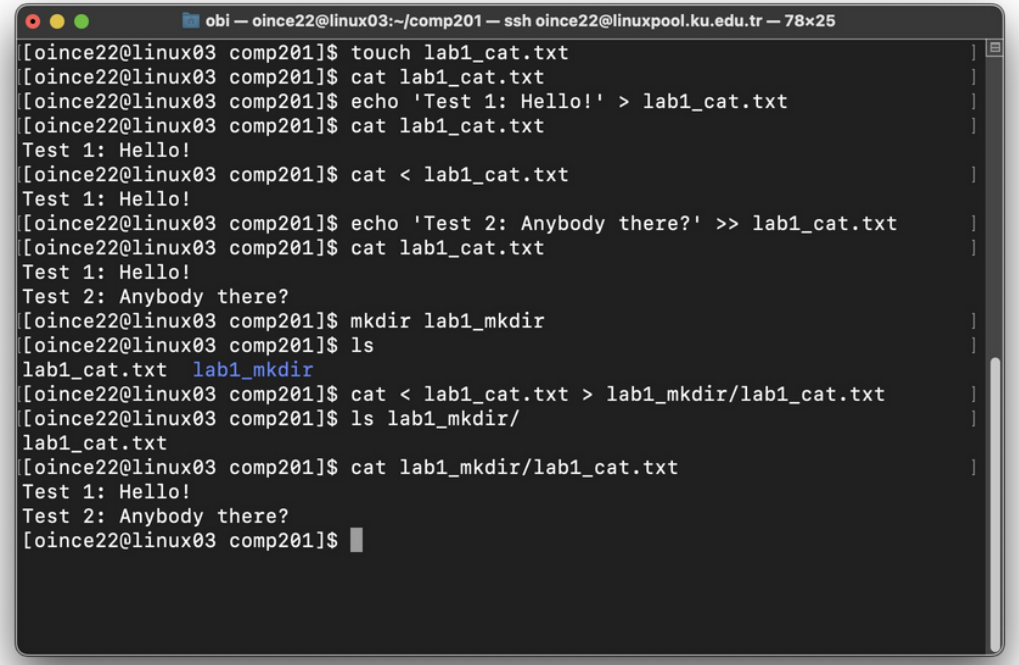
- To switch from normal mode to insert mode, type i in the normal mode.
- Every character you type is put to the file.
- To switch back to normal mode, press <Esc>

Operation Modes in Vim



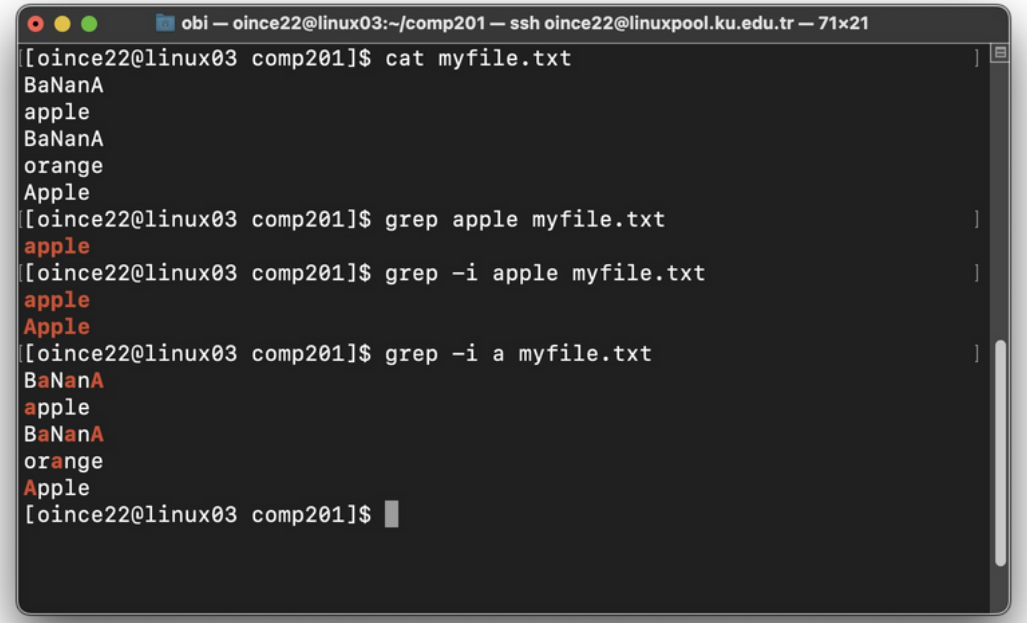
- Exit with saving
  - Tosave andexit a file, go to the Normal mode by pressing <Esc> then type :wq
- Exit without saving
  - To exit from afile without saving it, go to the Normal mode by pressing <Esc> then type :q!
- After typing :wq or :q!, press <Enter>

Redirection



- cat
  - Print the content of the given file
- < file and > file
  - You can write the input and output of a program to a file
  - ">> file" appends to end of file

Piping



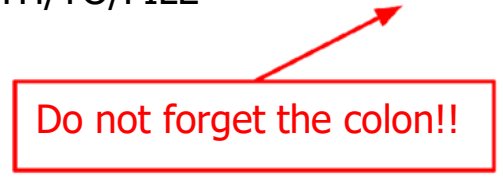
- Pipe character is |
  - Connects output of a program to input of another one
- grep
  - Searches for a particular information
  - By default it is case sensitive
- Try grep --help and find what does -i option do

SCP

- SCP is a tool in Linux used to transfer files between hosts over a network.
- The syntax for SCP is as follows:
  - scp [OPTIONS] SOURCE DESTINATION
- -r flag is used to copy directories, stands for recursive

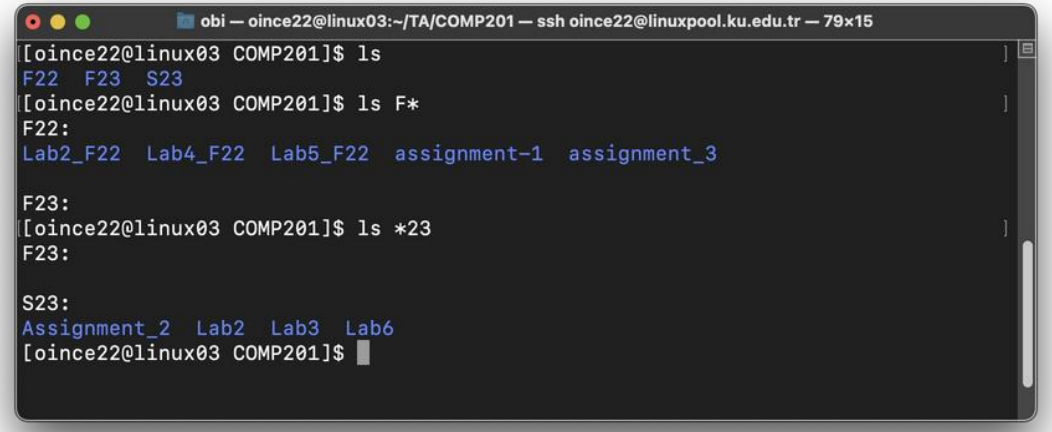
SCP

- From local machine to Linuxpool:
  - (on local machine): scp -r FILENAME USERNAME@linuxpool.ku.edu.tr:
- From Linuxpool to local machine:
  - (on local machine): scp -r USERNAME@linuxpool.ku.edu.tr:PATH/TO/FILE .



Useful Commands

- clear: Clearing the contents of the terminal screen
- history: Searching for previously executed commands
- Tab key: auto-completion
- * (asterisk): Used as a wildcard to represent any combination of characters in a command or filename



AI Agents

- An AI agent uses tools to complete a task.
- With access to your project, it can read files, edit code, and run terminal commands.
- It checks the results and decides what to do next.
- Examples: Cursor Agent and Claude Code.

AI Agents for Linux Practice

- Explain unfamiliar commands and flags.
- Investigate errors and automate repetitive file operations.

Mini demo: Permission denied

Running ./hello.sh gives "Permission denied".
Inspect the file and its permissions. Explain the cause,
apply the smallest necessary fix, and run it again.

Other Resources

- MIT MS The Shell
- Stanford CS107 Unix videos 1-15, 24, 25
- UNIX Tutorial for Beginners

Contributors:

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